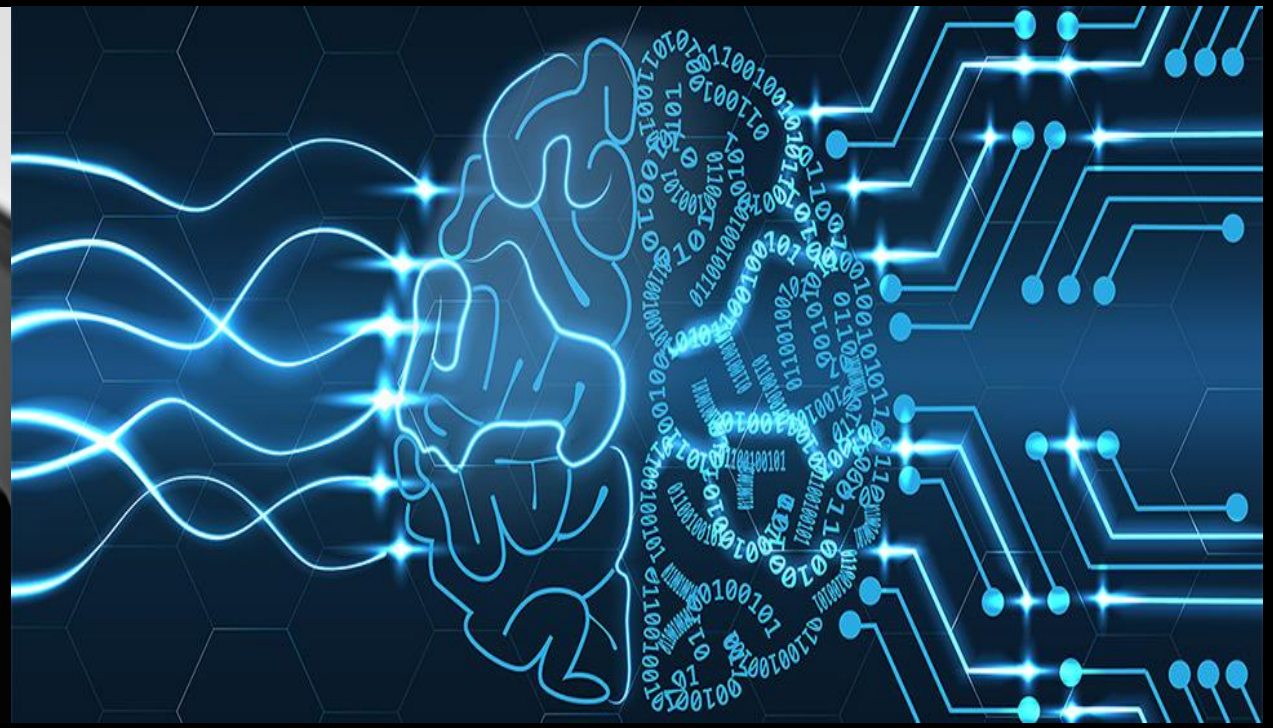
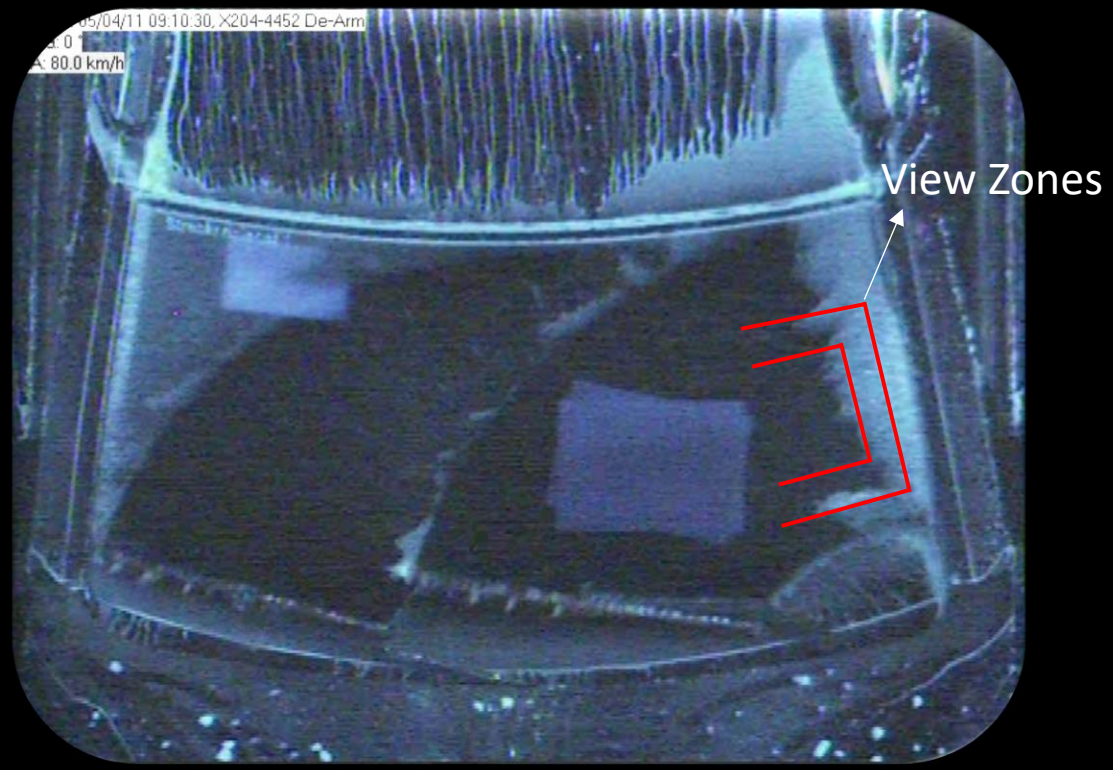


Wiper Performance Evaluations using Data Science

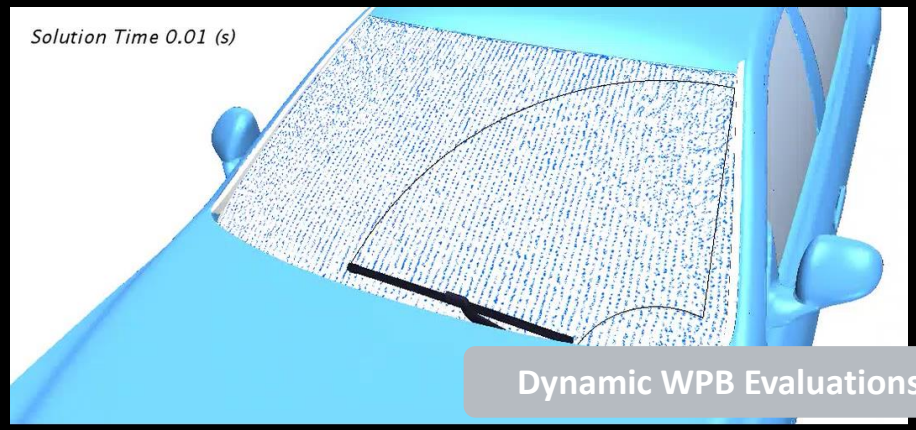
MBRDI Bangalore

Mercedes-Benz
The best or nothing.

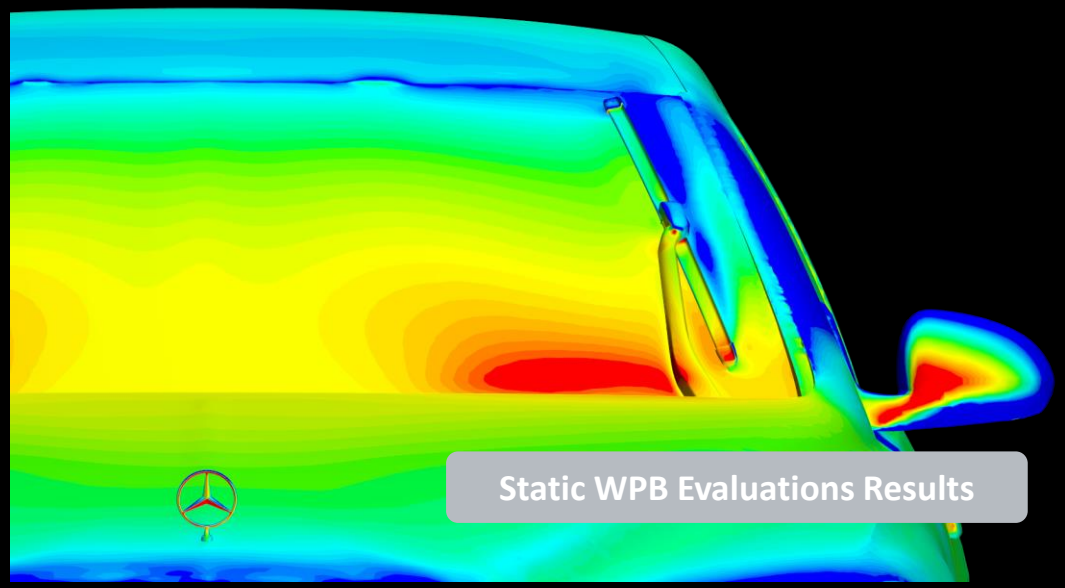
Introduction



Water Pullback Issue

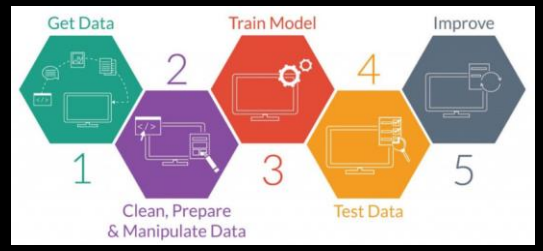


Dynamic WPB Evaluations Results

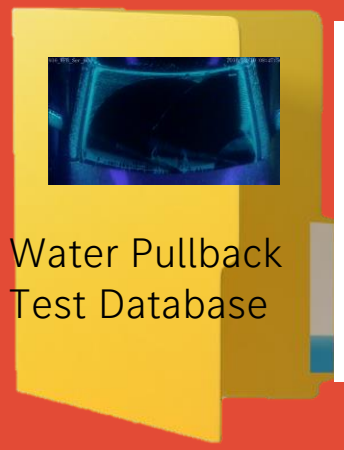


Static WPB Evaluations Results

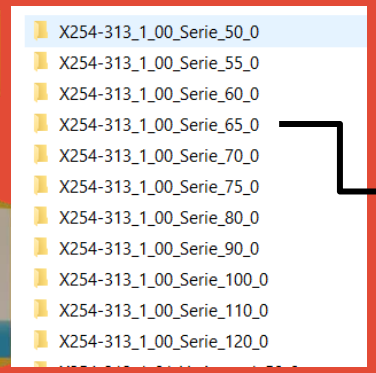
Proposed Methodology using Image Processing



Step 1: Training



Water Pullback
Test Database



Water Pullback Videos

+

Corresponding Wiper CAD

Inputs

Image Processing Algorithm

Learning...

Step 2: Prediction

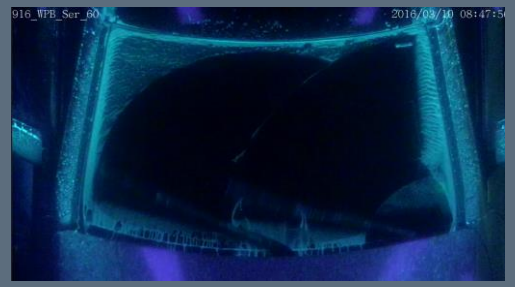
New Wiper CAD

Input

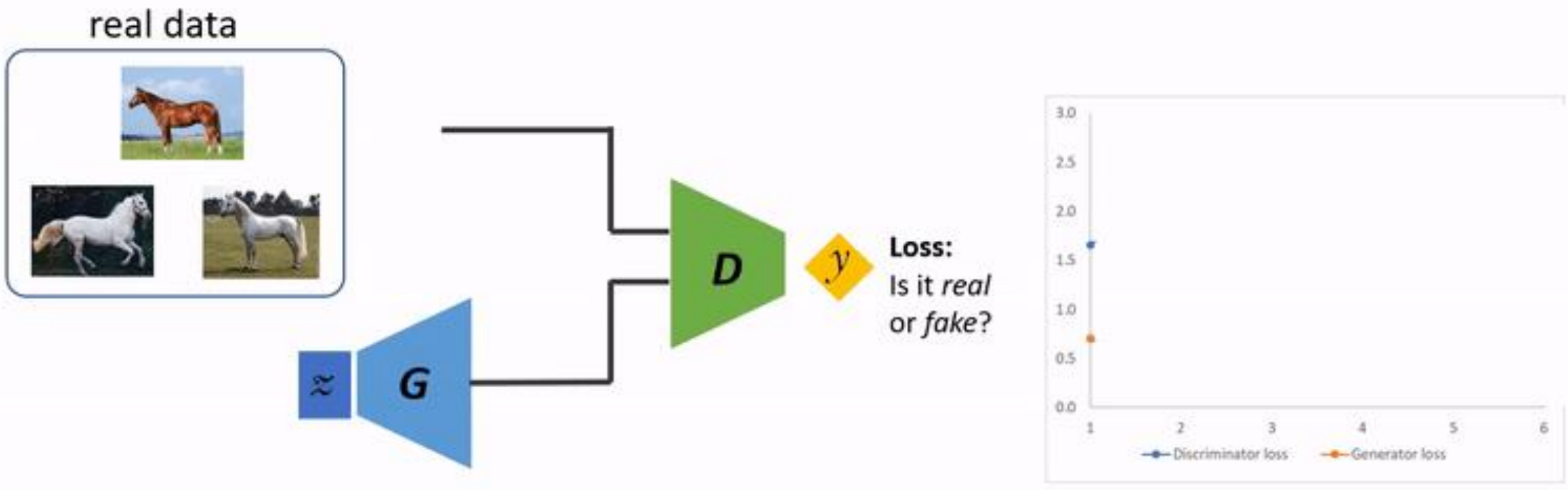
Image Processing Algorithm

Predictor

Output



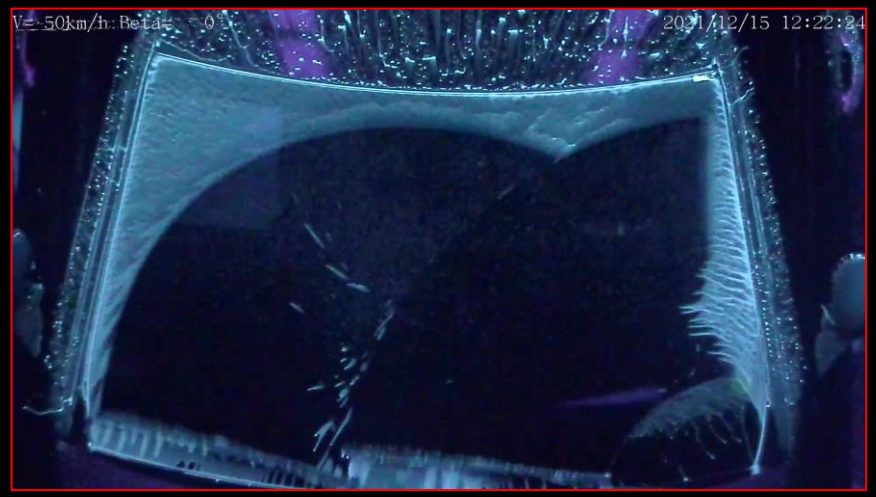
GANs Model



Note:
In GANs, the generator tries to predict a good image, and the discriminator tries to identify that image as fake. They both compete with each other, eventually forcing the generator to create correct images, and the discriminator to get confused between the real and generated image.

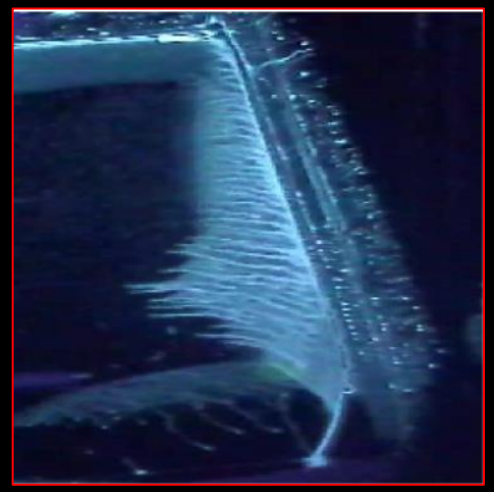
Wiper Pullback Image Generation – Preparation of Input

INPUTS: The data for the training of the model was prepared from 3 car-lines, the 206 model, the 214 model and the 254 model. The pullback images at different velocity timestamps was extracted from the videos and preprocessed as follows:



Initial image for velocity = 50kmph, 206 design

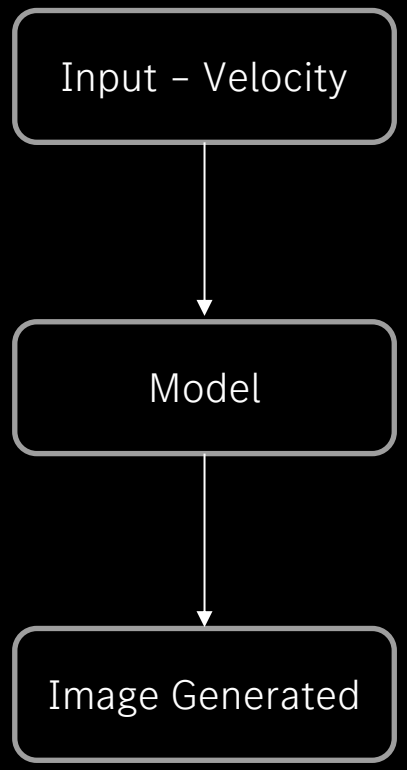
Extracted Pullback Info

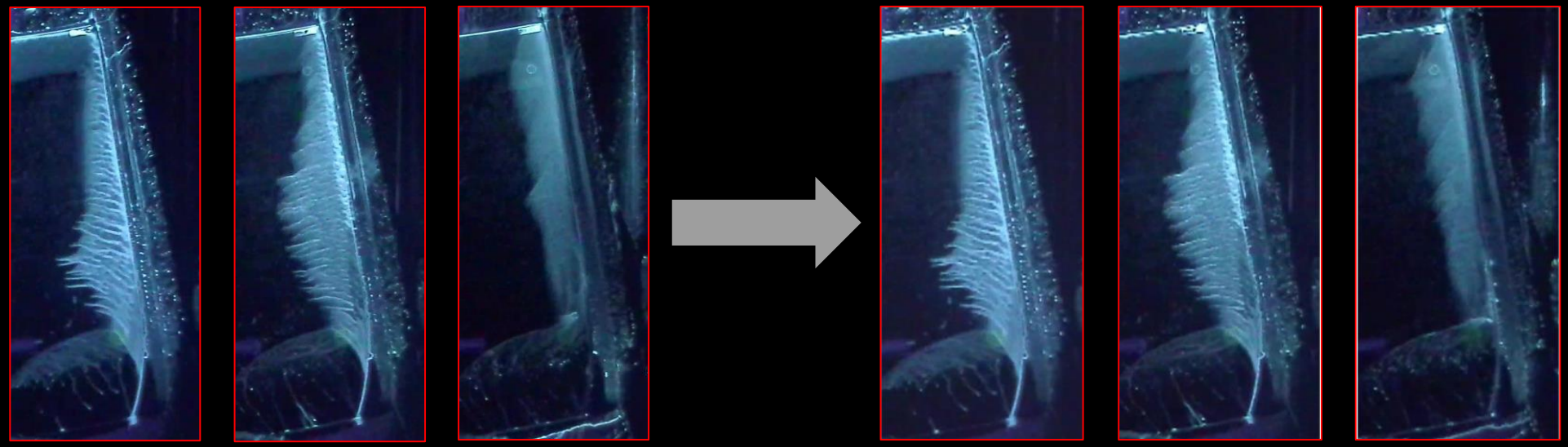


Reshaped image to 256x256 size to easily pass through our model

Preprocessing and extracting info pipeline

Wiper Pullback Image Generation – Velocity Extrapolation





Left – Original pictures of Pullback at 50, 65 and 90 kmph respectively, for the 206 design

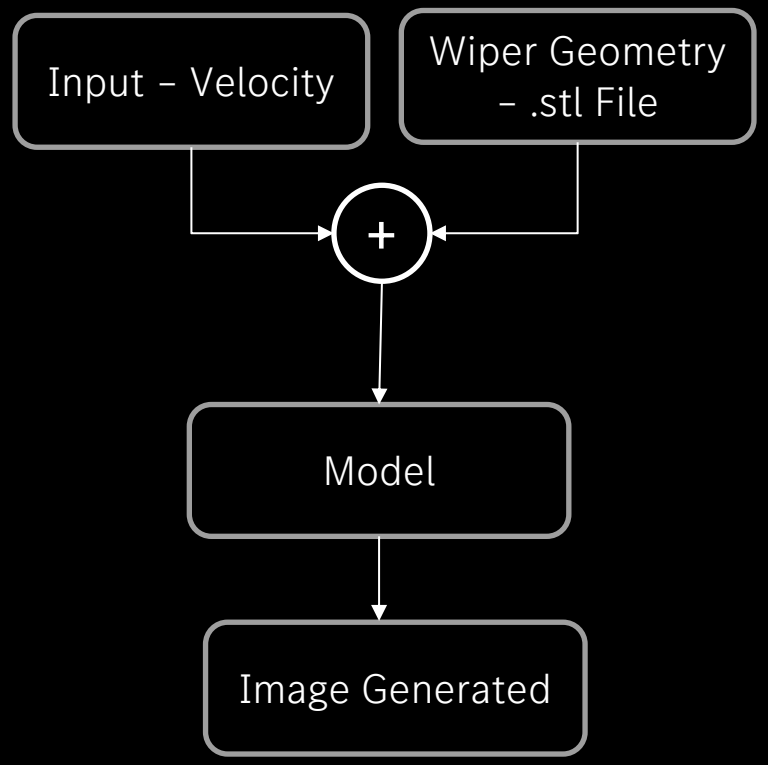
Right – AI Generated pictures of Pullback at 50, 65 and 90 kmph respectively, for the 206 design

Velocity Input is [50, 55, 65, 70, 75, 80, 90, 100, 110]kmph

Note:

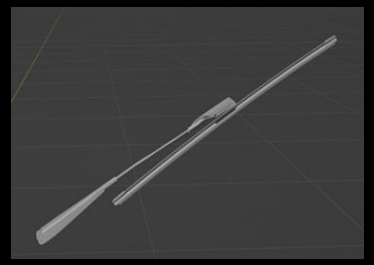
- The model accurately predicted variations in velocity, in both shape and size of pullback.
- As the model input starts from 50 kmph as the input it does not work for inputs less than 50 kmph

Wiper Pullback Image Generation – Design and Velocity Extrapolation

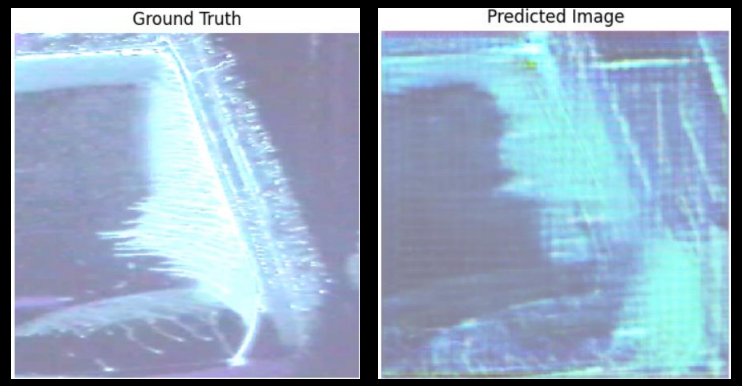


Inputs

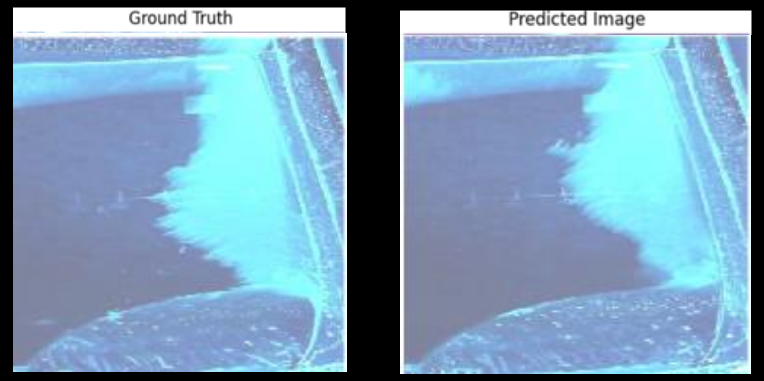
```
v_206 = [50, 55, 65, 70, 75, 80, 90, 100, 110]
v_214 = [40, 50, 60, 70, 80, 90, 120, 160]
v_254 = [50, 60, 80, 90, 100, 110, 120]
```



Ground Truth vs AI Generated Data



Design 206 – Velocity 50 kmph



Design 254 – Velocity 80 kmph

Note:

- It was observed that the model could not perform with great accuracy on the 206 model, while the 214 and the 254 models had high resolution predictions, but were still a bit off but the spread was more or less accurate
- The model struggled due to lack of design data for extrapolation, only 3 design data.

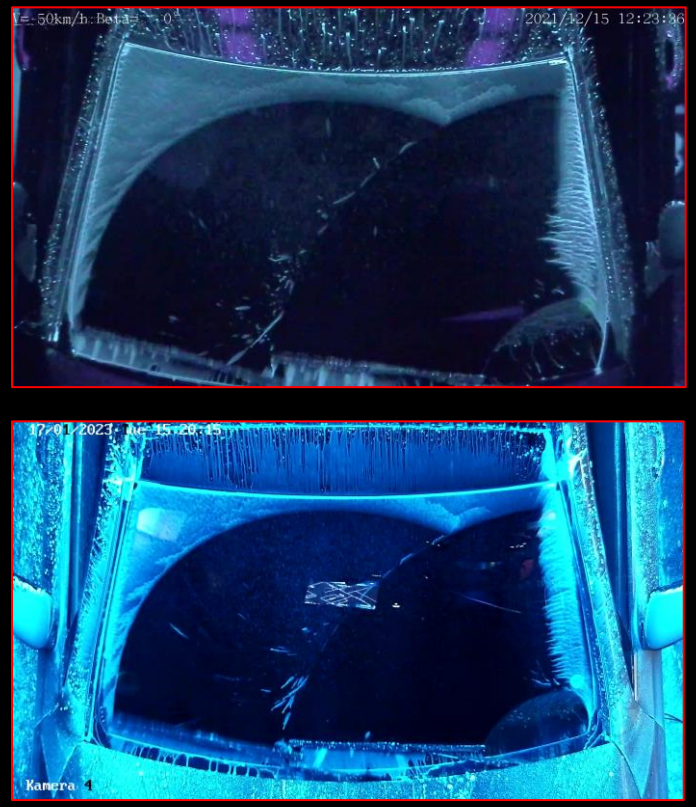
Wiper Pullback Image Generation: Challenges

Challenges Faced:

1. Data Limitations: Design files data outweighed velocity data in size of 1 datapoint, whereas variation in the amount of design data was very low (only 3 variants, all similar)
2. Lack of Uniformity: Training output had different type of images, hence confusing the model further

Current Limitations:

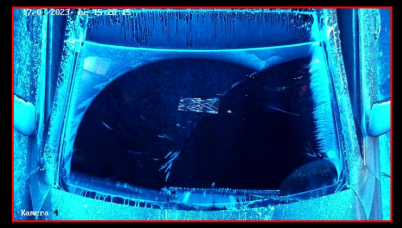
1. Insufficient design data remains a significant roadblock and GANs exhibit intricate learning patterns, warranting continuous tuning.



Hue difference between provided images of 206 model (up) and 254 model (down) for same velocity of 60 kmph tunnel wind

Wiper Pullback Image Generation: Requirements

Sifi



Resposibility: BTVs

- More data points(Test Videos/Images) from pullback evaluation of different designs.
- Wide variety of carline test data i.e from sedan, suv and hatchback.
- Minimum of 40 data points needed.

Folder with Videos

- X244
- W214 ...

Wiper Configurations Used

ToDos:

1. How soon 40 data points can be available? With corresponding CAD.

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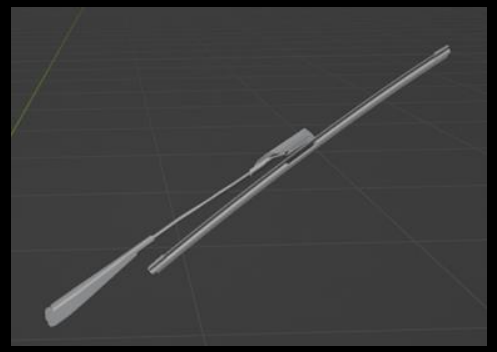
Responsibility: Ankita

- Model Development.
- MVP – For pullback prediction.

MBRDI Pune

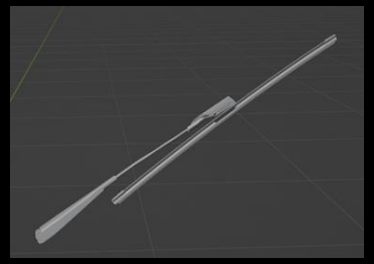


Responsibility: Mandar/Mayuri

- Identify the arrangement number for wiper configuration of the CAD.
- Share the Geometry Data

Wiper Pullback Image Generation: Requirements

254-270_1_00_Tageslicht_0_0.mp4	1/24/2022 1:30 PM
254-270_1_01_WPB_253_Wischer_65_0.mp4	1/24/2022 4:00 PM
254-270_1_01_WPB_253_Wischer_70_0.mp4	1/24/2022 4:01 PM
254-270_1_02_WPB_S_Schlag_X_Stellung_65_...	1/24/2022 6:02 PM



Carline	Wind Tunnel Test	Corresponding CAD Snapshot	Variant